Harsh

1.

Query : Where did the teacher explain `deadlock` ?
Upload: lecture notes and textbook PDFs
Result: OS_notes.pdf, page 12, and OS_book.pdf, page 340, with the word highlighted

2.

Home: “Where is that bill?”

Upload: electricity bills, bank statements, receipts

Search: `invoice 4821` or `warranty`

Result: the exact bill and page

Benefit: a personal document finder

3 .

Scenario: 10 years of annual reports, each 200 pages, full of tables.

Query: `net profit 2022`

What the engine does:

Extracts tables separately and stores each row as “Metric: Net Profit, Year: 2022, Value: 4,520 Cr”.

4 .

Faculty lecture notes and slides library

Who uploads: professors, one PDF per lecture.

Query: `process scheduling round robin`

Result: notes from several faculty members, with file name, page, and subject.

Extra feature: a `faculty_name` column, with an option to filter by professor.

Benefit: one search box across all subjects instead of hunting through WhatsApp groups and drives.

5 .

Search: `stack`

Results grouped by file:

DSA_slides.pdf (12 hits): stack data structure

OS_notes.pdf (5 hits): stack in process memory

CN_notes.pdf (3 hits): protocol stack

Benefit: the student sees one word means different things in different subjects and picks the relevant one. The filter by subject narrows it to one.

6.

Search: `kubernetes`

Result: “No results in the college library. Try: *docker*, *container*” or a note saying the topic is not in the uploaded material.

Why this matters: an honest empty result is better than irrelevant pages.

=========Keshaw Tiwari======================

Use cases for Search Engine (Emphasis vector matching)

Expected usage:

Explain the Longest Integer Subsequence formulae from dynamic programming :-> returns a summary or bigger document (depending on the need)

Summarization capability:

2.1 Text: Example: Summarize CIDR from Computer Networks

2.2 Video: Example: Explain in brief paging from Memory Management video

Procedure: Locates the timestamps covering paging, extracts related transcript

Treats same as Text

2.3 Audio: Same treatment as video

List related topics:

For example its imperative to study graphs, recursion while studying dynamic programming because dp problems often require to be modelled with a graph.

Create new stuff from available content: (Indirect usage)

Useful for teachers: Create a quiz on frequency filtering from Image Processing slides

Procedure: Matches keywords “Frequecy filtering” and “Image Processing” with relevant content which can be later used to create quizzes .

Browsing related content: The engine can track the user and build up a history which can be used to display related content. Its beneficial to index calculus related content while studying sequence and series.

Knowledge graph: Related to #5, Organizes related content into a Directed Acylic Graph

(T1) -> (T0) -> (T3)

For example a user wanting to study topic T0 would need studying T1 and after studying T0 he may study T3.

Related to #6, slightly more ambitious idea, topics belonging to the same content blob are more easily matched, but to increase capabilities of the knowledge graph we connect chunks:

For example ideas of space and time complexity from data structure are highly correlated with Sustainability topics, data centers need to be efficient, can figure out these complex relationships.

Internet connectivity with the search engine: Can make use of advance google search on topics.

Topic statistics: Can track most viewed topic

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Riya Sharma

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Use Cases for Search Engine:

1. Queries with restriction:

For example: Explain overfitting using linear regression, not neural networks.

Now: to solve it Engine uses constant extraction to identify what student wants and what they wants to avoid.

Then by reranking model we can check whether the retrived chunks follows these conditions.

2. Where Two concepts are compared:

Sometime students remembers that two concept were compared but cannot remember the exact section.

Student ask:

“Where are process and threads compared,including their memory usage and context switching differences?”

Then Engine Uses semantic vector retrieval and comparison focused query decomposition to retrieve chunks using both concept and their differences then by prioritizes chunks that directly compare the requested properties.

Benefit: Student find direct comparison instead of collecting Separate definition manually.

3. Sometimes student encounters difficult problem and wants to identify concept needed to solve it.

For example student ask:

“what concept should I know before solving a longest increasing subsequence problem?”

What engine does: Uses concept extraction and prerequisite mapping to identify concepts required by relevant explanations and worked solutions. It can derive prerequisite relationships from the study material and validate them against the retrieved content before presenting the results.

4.Study materials contain technical abbreviations and symbols that are difficult to retrieve through ordinary text search.

What the engine does: Uses query normalization, abbreviation expansion, and hybrid retrieval. Common abbreviations are mapped to their expanded forms, while mathematical symbols and technical notation are preserved during indexing.

5. Problem: Sometimes a student follows a solution but gets a different answer and cannot understand where they made the mistake.

What the engine does: Uses semantic search and step-by-step comparison to find similar solutions from the study material. Then, it compares the student's solution with the retrieved solution to find where the mistake might have happened, such as updating the wrong variable or missing a condition.

========= Narmata Goel ==========

1. Finding Formulas from Natural Language Queries

Scenario: Students remember the concept behind a formula but forget the actual mathematical expression.

Query: “What is the formula for force?”

What the engine does:

- Uses semantic search to find explanations related to Newton's second law.
- Retrieves the relevant formula, (F = m \times a), along with its explanation.
- Ranks the most relevant information higher in the search results.

Benefit: Students can find formulas even when they do not remember the exact terminology used in their textbooks.

2. Finding Information Using Real-World Descriptions

Scenario: Students understand a concept but do not know its technical name.

Query: “How do I calculate the space inside a round shape?”

What the engine does:

- Matches the query with relevant geometry concepts.
- Retrieves the explanation and formula for the area of a circle, (A = \pi r^2).
- Includes related definitions to help the student understand the result.

Limitation: A round shape could refer to a circle, sphere, or cylinder.

Possible improvement: Use context to identify the intended shape or ask the user to clarify.

3. Solving Numerical Problems Using Retrieved Knowledge

Scenario: Students need to apply formulas from their study materials to solve numerical problems.

Query: “Calculate simple interest for ₹5,000 at 6% per annum for 2 years.”

What the engine does:

- Retrieves the simple interest formula from the relevant study material.
- Identifies the principal, interest rate, and time from the query.

- Passes the formula and values to a calculation module.

Result: Simple interest = ₹600.

Benefit: Combines information retrieval with calculation to help students solve practical problems.

4. Answering Questions That Require Information from Multiple Sections

Scenario: The complete answer to a question is distributed across different sections of a textbook.

Query: “What does a plant need to make food, and what is this process called?”

What the engine does:

- Retrieves information explaining photosynthesis.
- Finds related information about sunlight, water, and carbon dioxide.
- Combines the retrieved content into a complete answer.

Limitation: Retrieving only one relevant section may produce an incomplete answer.

Benefit: Students receive a complete explanation without manually searching through multiple sections.

5. Searching Mathematical Concepts Using Spoken or Informal Notation

Scenario: Students remember how a formula sounds but cannot write its mathematical symbols correctly.

Query: “What does sigma x divided by n mean?”

What the engine does:

- Recognizes the mathematical expression (\sum x/n).
- Retrieves information about the arithmetic mean.
- Explains the meaning of the summation symbol and the number of observations.

Benefit: Makes mathematical information easier to find for students who remember formulas verbally rather than symbolically.

6. Discovering Formulas Through Related Technical Terms

Scenario: Students remember the quantities involved in a formula but forget the name of the law.

Query: “How are amps, volts and ohms related?”

What the engine does:

- Identifies the relationship between electric current, voltage, and resistance.
- Retrieves the explanation of Ohm's law and the formula (V = I \times R).
- Displays the variable definitions and measurement units.

Benefit: Students can discover relevant concepts even when their query does not contain the exact technical name.

7. Comparing Concepts and Formulas Across Subjects

Scenario: Students want to compare concepts explained in different chapters or subjects.

Query: “Compare Newton's second law with Ohm's law.”

What the engine does:

- Retrieves the formula for force, (F = m \times a).
- Retrieves Ohm's law, (V = I \times R).
- Finds their explanations and variable definitions.
- Organizes the results into a side-by-side comparison.

Benefit: Students can understand similarities and differences without manually searching for each concept separately.